

SDA Vision / Research report

earth-image.jpg



Assessment summary

Field	Recorded detail
Finding	Leaning authentic
Recorded verdict	authentic likely
Confidence	medium
Indicative rating	4/100
Rating basis	median of available model ratings; not a calibrated probability
Explanation	3 models lean authentic with matching verdicts and ratings. Review the other findings, original source and context alongside this result.
Coverage	image - 1 frame(s) analysed
Review context	Research assessment. Review alongside source information and context. When credentials are missing, origin remains unresolved by this check. Use these findings to inform a documented human review.

Provider overview

Provider	Type / status	Assessment	Rating
Claude Vision	vision / ok	authentic likely	4/100
OpenAI Vision	vision / ok	authentic likely	2/100
Gemini Vision	vision / ok	authentic likely	25/100
C2PA Content Credentials	provenance / ok	inconclusive	n/a
Local forensic cues	forensic / ok	partially synthetic	47/100

Provider evidence / 1

Field	Recorded detail
Provider	Claude Vision
Model	claude-opus-5-5
Status	ok
Assessment	authentic likely
Confidence	high
Role in assessment	Supporting
Provider summary	synthetic likelihood 4/100; tells: visually consistent with the widely documented Apollo 17 'Blue Marble' photograph (NASA AS17-148-22727, Dec 1972), geographically accurate coastlines: Africa, Arabian Peninsula, Red Sea, Gulf of Aden, Madagascar, Antarctic ice cap, physically coherent cloud systems and cyclonic swirls with plausible meteorological structure, film-scan grain and slight color cast consistent with analog photography
Evidence 1	Visually consistent with the widely documented Apollo 17 'Blue Marble' photograph (NASA AS17-148-22727, Dec 1972)
Evidence 2	Geographically accurate coastlines: Africa, Arabian Peninsula, Red Sea, Gulf of Aden, Madagascar, Antarctic ice cap
Evidence 3	Physically coherent cloud systems and cyclonic swirls with plausible meteorological structure
Evidence 4	Film-scan grain and slight color cast consistent with analog photography
Evidence 5	Natural limb shading and terminator-free full-disc illumination consistent with sun behind the camera
Evidence 6	No repeated textures, smeared detail, or generative blending artifacts at land/cloud boundaries
Evidence 7	Caveat: Visual match to a famous archival image is not proof of an unaltered copy; subtle retouching, recoloring, or a high-fidelity regenerated replica cannot be ruled out without comparison to the NASA source scan and metadata/provenance checks.

Provider evidence / 2

Field	Recorded detail
Provider	OpenAI Vision
Model	gpt-6-astra
Status	ok
Assessment	authentic likely
Confidence	high
Role in assessment	Supporting
Provider summary	synthetic likelihood 2/100; tells: Coherent African, Arabian Peninsula, and Madagascar coastlines, Intricate, nonrepeating cloud bands and storm spirals, Consistent atmospheric haze and foreshortening toward Earth's limb, Fine grain and slight softness consistent with a scanned photograph
Evidence 1	Coherent African, Arabian Peninsula, and Madagascar coastlines
Evidence 2	Intricate, nonrepeating cloud bands and storm spirals
Evidence 3	Consistent atmospheric haze and foreshortening toward Earth's limb
Evidence 4	Fine grain and slight softness consistent with a scanned photograph
Evidence 5	Composition closely resembles the historical Blue Marble photograph
Evidence 6	Caveat: Visual resemblance cannot establish provenance; compare with the archival original to check for subsequent edits.

Provider evidence / 3

Field	Recorded detail
Provider	Gemini Vision
Model	gemini-3.8-flash
Status	ok
Assessment	authentic likely
Confidence	medium
Role in assessment	Supporting
Provider summary	synthetic likelihood 25/100; tells: authentic film grain, consistent atmospheric limb scatter, historically documented cloud formations, natural optical sensor characteristics
Evidence 1	Authentic film grain
Evidence 2	Consistent atmospheric limb scatter
Evidence 3	Historically documented cloud formations
Evidence 4	Natural optical sensor characteristics
Evidence 5	Caveat: High-profile historical images can be digitally remastered or simulated by generative models.

Content Credentials / provenance

Field	Recorded detail
Provider	C2PA Content Credentials
Model	Not recorded
Status	ok
Assessment	inconclusive
Confidence	Not recorded
Role in assessment	Additional finding
Provider summary	Embedded credentials: none found. Origin remains unresolved by this check.
Evidence 1	Coverage: credentials embedded in the submitted file. External credentials were not retrieved.
Evidence 2	Verifier: c2pa-python 0.37.12; native SDK 0.91.0; specification target 2.4; policy embedded-offline-reviewed-trust-v3.
Evidence 3	Offline embedded-manifest verification; no network trust-list, manifest or revocation refresh.
Evidence 4	No reviewed local C2PA trust snapshot installed. Provenance overrides are disabled.

Provider evidence / 5

Field	Recorded detail
Provider	Local forensic cues
Model	Not recorded
Status	ok
Assessment	partially synthetic
Confidence	low
Role in assessment	Additional finding
Provider summary	Heuristic synthetic likelihood 47/100 from metadata + pixel statistics.
Evidence 1	No camera EXIF (common in AI exports, screenshots, and re-saves).
Evidence 2	ELA mean 0.5, peak 10 (uniform high ELA can indicate generation/heavy edit).
Evidence 3	High-frequency noise std 4.61 (very low/very uniform noise is a synthesis cue).
Evidence 4	Spectral peakiness 25.63 (periodic frequency peaks can mark generated textures).

Evidence supporting this verdict

Field	Recorded detail
Claude Vision	synthetic likelihood 4/100; tells: visually consistent with the widely documented Apollo 17 'Blue Marble' photograph (NASA AS17-148-22727, Dec 1972), geographically accurate coastlines: Africa, Arabian Peninsula, Red Sea, Gulf of Aden, Madagascar, Antarctic ice cap, physically coherent cloud systems and cyclonic swirls with plausible meteorological structure, film-scan grain and slight color cast consistent with analog photography
OpenAI Vision	synthetic likelihood 2/100; tells: Coherent African, Arabian Peninsula, and Madagascar coastlines, Intricate, nonrepeating cloud bands and storm spirals, Consistent atmospheric haze and foreshortening toward Earth's limb, Fine grain and slight softness consistent with a scanned photograph
Gemini Vision	synthetic likelihood 25/100; tells: authentic film grain, consistent atmospheric limb scatter, historically documented cloud formations, natural optical sensor characteristics

Additional findings

Field	Recorded detail
Local forensic cues	Supporting observation; considered alongside the model assessments.
C2PA Content Credentials	Embedded credentials: none found. Origin remains unresolved by this check.

How this assessment was reached

Field	Recorded detail
1. override	outcome=none; detail=No confirmed C2PA AI assertion or SynthID watermark. Origin remains unresolved by these checks.
2. model_agreement	synthetic=[]; authentic=['Claude Vision', 'OpenAI Vision', 'Gemini Vision']; uncertain=[]; forensic_supports_synthetic=False
3. verdict	value=authentic_likely; confidence=medium; reason=3 models lean authentic with matching verdicts and ratings. Review the other findings, original source and context alongside this result.

Cite this analysis

Field	Recorded detail
Software (Harvard)	Martin, S. (2026) SDA Vision: Synthetic-media Discourse Analysis (version 0.1.0) [Computer software]. Manchester Metropolitan University. Available at: https://sdavision.io/
Software creator	Dr Sam Martin, Smart Data Research UK (UKRI) Fellow, Manchester Metropolitan University (MMU). ORCID: https://orcid.org/0000-0002-4466-8374 .
Funding acknowledgement	This work was supported by Smart Data Research UK, a UKRI investment; Grant number UKRI4010.
Analysis record	File: earth-image.jpg; media type: image; recorded at: 2026-09-26T22:31:03+00:00; models: claude-opus-5-5, gemini-3.8-flash, gpt-6-astra.

File and analysis record

Field	Recorded detail
File	earth-image.jpg
Analysis ID	afa06964-dc8c-4c81-8fc7-b6179d9b2705
Generated	2026-09-26T22:31:03+00:00
Tool	SDA Vision 0.1.0
Models	claude-opus-5-5, gemini-3.8-flash, gpt-6-astra
Method version	pdf-preparation-2026-09-26.1
Analysis lens	general
Original SHA-256	a4bad2aba040fd992aff7483eee6166943e232061b82d7149b2060786719753b
Ingestion note 1	Fresh provider replies acquired for eleven-examples-refresh-20260926-r1; report assembled locally from the saved responses without further API calls.
Ingestion note 2	Provider latency values are restored from the acquisition receipts (maximum sampled-frame latency per provider); raw replies and assessment results are unchanged by this bookkeeping step.